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Review article

Biomolecular condensates in plant cells: Mediating and integrating environmental signals and development

Yang Huang, Pengguo Xia *,[1]

College of Life Sciences and Medicine, Zhejiang Sci-Tech University, Hangzhou 310018, China

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ABSTRACT

In response to the spatiotemporal coordination of various biochemical reactions and membrane-encapsulated organelles, plants appear to provide another effective mechanism for cellular organization by phase separation that allows the internal compartmentalization of cells to form a variety of membrane-less organelles. Most of the research on phase separation has centralized in various non-plant systems, such as yeast and animal systems. Recent studies have shown a remarkable correlation between the formation of condensates in plant systems and the formation of condensates in these systems. Moreover, the last decade has made new advances in phase separation research in the context of plant biology. Here, we provide an overview of the physicochemical forces and molecular factors that drive liquid-liquid phase separation in plant cells and the biochemical characterization of condensates. We then explore new developments in phase separation research specific to plants, discussing examples of condensates found in green plants and detailing their role in plant growth and development. We propose that phase separation may be a conserved organizational mechanism in plant evolution to help plants respond rapidly and effectively to various environmental stresses as sessile organisms.

1. Introduction

The cell is the smallest functional unit that makes up an organism. It separates itself from the external environment using a membrane to ensure the internal reaction system is not disrupted. During the long evolutionary process, to carry out specific biochemical reactions at particular sites, the interior of eukaryotic cells has also evolved several membrane-encapsulated organelles and compartments. Examples of membrane-encapsulated organelles in eukaryotes include lysosomes, endoplasmic reticulum, and Golgi apparatus. These organelles are isolated from their surroundings by membranes, and different cellular metabolism and signaling pathways can occur within them. For example, the nucleus segregates transcription from translation in physical space to facilitate the development of a posttranscriptional control system(Glisovic et al., 2008). In addition to the classical membrane-bound organelles, a variety of membrane-free compartments exist in cells that perform complex biochemical functions by concentrating on specific liquid-like components, such as enhancing activity through focusing enzymes and substrates, regulating macromolecule folding state, regulation of mRNA translation and deadenylation (Guillen-Boixet et al., 2020; Kim et al., 2019; Pak et al., 2016; Sheu--Gruttadauria and MacRae, 2018; H. Wang et al., 2019). These membrane-free blocks have been described in the literature under different terms, including membrane-less organelles, droplets, granules, speckles, and aggregates, which we call biomolecular condensates (Table 1).

It is now generally accepted that in organisms, liquid–liquid phase separation (LLPS) is involved in the formation of biomolecular condensates. These condensates are engaged in one or more cellular reactions in cells, but various factors regulate their exact mechanisms and need to be further explored.(Alberti and Hyman, 2021; Kusumaatmaja et al., 2021; Sabari et al., 2020). Here, we review the general concept of LLPS in all living systems and its basic biophysical properties and molecular knowledge. Next, we introduce representative condensates in plants based on the structure of the cell and illustrate how plants utilize the reversible properties of condensates to respond to endogenous and external environmental stimuli. Finally, we discuss the potential of phase separation in plants and the unique challenges plants may face.

* Corresponding author.
E-mail address: xpg_xpg@zstu.edu.cn (P. Xia).
[1] Orcid ID: 0000–0003-3572–7616

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Table 1
Biomolecular condensates in plants.

Localization	Biomolecular Condensates	Function	Refs
Nucleus	Nuclear speckle	coordinate nuclear gene expression and play essential roles in splicing regulation, RNA metabolism, and gene expression regulation	(Alam et al., 2022; S. Xu et al., 2022)
	Photobodies	circadian rhythms, de-yellowing, senescence, flowering, geotropism, light perception, and temperature perception.	(Ronald and Davis, 2019)
	ELF3 condensates	involved in the expression of critical regulators of photosynthesis, the biological clock, phytohormone signaling, growth, and response to the environment.	(Jung et al., 2020; H. Zhao et al., 2021)
	GDACs	acts as a plant defense mechanism to enhance plant disease resistance and immunity.	(S. Huang et al., 2021)
	FCA condensates	RNA-binding protein that interacts with processing elements at the 3′-end of RNA and plays a vital role in the autonomous pathway of flowering.	(Fang et al., 2019)
	Nucleolus	ribosome construction, rRNA biosynthesis, and stress defense in plants.	(Kalinina et al., 2018)
	D-bodies	Processing of Pri-mirna into mature miRNAs	(Q. Li et al., 2021)
	Cajal bodies	RNA-mediated gene silencing, mRNA metabolism.	(Ohtani, 2017)
Cytoplasm	TMF condensates	binding to the promoter regulates flowering time.	(X. Huang et al., 2021)
	ARF condensates	involved in the growth hormone transcriptional response.	(Powers et al., 2019)
	SINCs	involvement in plant defense mechanisms induced by SA.	(Raul Zavaliev and Dong, 2023; R. (Zavaliev et al., 2020)
	FLOE condensates	involved in water potential sensing in embryos in seeds.	(Dorone et al., 2021)
	Stress granules	inhibition of translational activity during the stress response.	(Kosmacz et al., 2019)
	Processing bodies	enhanced stress response, involved in RNA processing and mRNA degradation.	(Decker and Parker, 2012; Pomeranz et al., 2010)
	LEA condensates	involvement in drought tolerance as well as plant stress mechanisms.	(Cuevas-Velazquez et al., 2016)
	siRNA bodies	siRNA production and gene silencing	(Tan et al., 2023)
Organelle	STT condensates	mechanisms of protein sorting in chloroplasts.	(Ouyang et al., 2020; X. Xu et al., 2021)
	Pyrenoid matrix	involved in CO_2 enrichment.	(S. He et al., 2023; Rosenzweig et al., 2017)

2. LLPS and its molecular features

LLPS is a highly dynamic, reversible cellular mechanism that allows a homogeneous mixed solution of macromolecules to spontaneously divide into a concentrated and dilute phase (Fig. 1A). The concentrated phase has liquid-like properties and is enriched with specific macromolecules, while other macromolecules are diluted in the process, allowing the focused phase to function as a compartment(Alberti et al., 2019). In addition, LLPS can form a droplet-like structure and continue to transform into a gel-like form. LLPS in the gel state is often irreversible. This also provides a new perspective on developing amyloid-like fibers in vivo(Leppert et al., 2023; Woodruff et al., 2018). It has also been shown that mutations in specific proteins can accelerate this process of LLPS to the gel state(Alberti and Hyman, 2021; Patel et al., 2015). The most critical factors affecting the LLPS process include two aspects: first, the macromolecular substances involved, such as proteins and nucleic acids, and second, the environmental factors, including pH, temperature, solution concentration, and post-translational modification (PTM) of proteins(Boeynaems et al., 2018; Chong and Forman-Kay, 2016). These factors play a crucial role in the LLPS process and have a decisive influence on the process and outcome of LLPS (Fig. 1B).

3. Molecular features driving biocondensate formation

Proteins involved in LLPS are usually intrinsically disordered proteins (IDPs), and intrinsically disordered regions (IDRs) are typically present within IDPs, and about one-third of eukaryotic proteins are IDPs (Chong and Forman-Kay, 2016; Elbaum-Garfinkle et al., 2015; Uversky, 2017). IDPs are a class of proteins that contain an unfixed three-dimensional structure(Li and Pang, 2023). This feature makes them highly variable and allows them to adjust their arrangement when bound to stationary molecules. The flexible conformation enables interactions between proteins containing multivalent properties. (Bah and Forman-Kay, 2016; Das et al., 2015). In addition, the ability of a molecule to participate in multiple intermolecular interactions determines the propensity for phase separation(Dao et al., 2018; X. Yang et al., 2024). In general, multivalent molecules have a more remarkable ability to facilitate phase separation, and proteins containing multiple folding module structural domains or IDRs can facilitate these multivalent interactions.(Gomes and Shorter, 2019; Holehouse and Kragelund, 2023). However, other biopolymers like DNA or RNA can also encode multivalency and interact(Pessina et al., 2021; Stanic and Mekhail, 2022; Udono et al., 2023; Y. L. Wang et al., 2023). The IDRs of many proteins are required for phase separation to occur in vitro, and a large number of IDR-containing condensates have been found to date in a variety of living systems; however, it is worth noting that IDR-containing proteins are not the only proteins that undergo phase separation(Babu, 2016; Kulkarni et al., 2022; Oldfield and Dunker, 2014). In addition, different IDRs have different functions, with some IDRs promoting phase separation and others inhibiting it.

The IDRs of IDPs usually have specific amino acid compositions and sequences, often referred to as low-complexity domains (LCDs), and play essential roles in cellular signaling, PTM, processing of intracellular

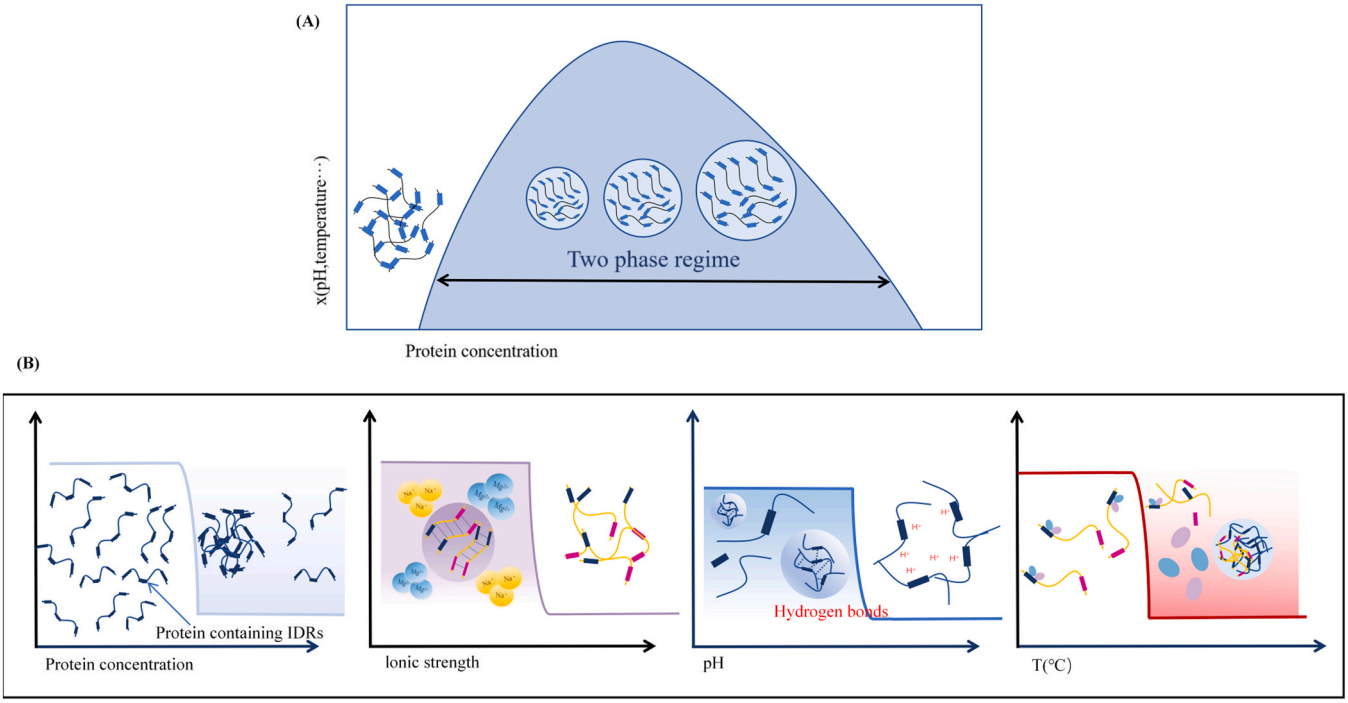

Fig. 1. Concentration dependence of LLPS and the forces driving LLPS. (A) The occurrence of LLPS in simple component systems, protein-protein two-component systems, depends on the concentration of protein and other factors, such as temperature and pH, among others. Under fixed conditions, as the concentration in the total system increases, the proteins of the single-phase system, which are uniformly distributed in the system, are transformed into concentrated and dilute phases, as well as the formation of condensates. (B) Relationship between coexistence curves of biomolecules and the interaction parameter *x*. Various environmental factors, including pH, temperature, ionic strength, IDRs, and others, influence condensate formation via LLPS by inducing interactions between biomolecules.

substances, and protein degradation(Cermakova and Hodges, 2023; Tsang et al., 2020; C. Xu et al., 2023). However, for LLPS, the precise sequence of LCDs is usually not paramount; the overall composition and charge pattern matter. The hydrophobic, aromatic ring and charged nature of amino acid residues are the basis for LCD-induced phase separation (Fig. 1B). LCD mediates multivalent interactions through a variety of interaction modes, including charge-charge, cation-πbonds, π-π stacking, hydrophobicity, and hydrogen bonding, and these multivalent interactions between biomolecules are the key molecular drivers for the formation of condensates(Fomicheva and Ross, 2021; Jang and Engelman, 2023; Peran and Mittag, 2020). A prominent class of LCDs is a prion-like domain (PrLD) containing polar uncharged amino acid residues that are typically found in RNA-binding proteins (RBPs). PrLD-containing proteins were initially recognized for their ability to assemble into self-templating protein aggregates(S. Jiang et al., 2020; Martin et al., 2020); Joshua A. (Riback et al., 2020). Increasingly, PrLD has been shown to have the ability to form condensates through phase separation. The FLOWERSHIP CONTROL LOCUS A (FCA), which contains PrLD, can regulate flowering through the LLPS, and FUS PrLD can undergo an LLPS driven by tyrosine-tyrosine interactions(Burke et al., 2015; Fang et al., 2019; Murray et al., 2017). In addition, PrLDs do not interact with other PrLDs during condensate assembly but rather with other IDRs, or modulate phase behavior by affecting the degree of protein solubility(J. Wang et al., 2018).

4. Biomolecular condensates in plants

Most of the available evidence supports the possibility that LLPS may underlie the formation of condensates centers on fungal systems and animals. However, these condensates can also be present in plants. Considering that LLPS is extremely sensitive to changes in physicochemical conditions, which suggests that phase separation may play an essential role in stress adaptation and that plants, as sessile organisms, are constantly subjected to a variety of external biotic and abiotic stresses from the developmental stage, the occurrence of LLPS facilitates the response of plants to changing environments under emergency conditions to improve the adaptive capacity of plants. This means that in plants, these condensates are similarly formed by LLPS. Some biocondensates common in plants and animal systems, such as Cajal bodies, stress granules, dicing body, P-bodies, etc., have been extensively reviewed(Q. Li et al., 2021; Taliansky et al., 2023; Tan et al., 2023). Therefore, we will follow the subcellular localization of various condensates to introduce plant-specific condensates formed through phase separation that are involved in various processes of plant growth and development and play crucial roles in plant response to various stresses including but not limited to temperature and rhythm receptors, plant defense mechanisms, sensing of external moisture to sprout in optimal environments, and so on.

4.1. Biomolecular condensates in the nucleus

4.1.1. Photobody

For plants, light changes from seed germination to flowering are one of the main ways for plants to acquire rhythmic information. In *Arabidopsis thaliana*, five classes of phytochromes are present: phytochromes A to phytochromes E (phy A-phy E), and phy B is the primary receptor for red and far-red light(Quail, 2021). Phytochromes are capable of sensing red and far-red light and play a regulatory role in various stages of plant development, including circadian rhythms, de-yellowing, senescence, flowering, and geotropism(Paik and Huq, 2019). Upon absorption of red/far-red light, phy B undergoes a reversible change in conformation, thus becoming a photosensitive pigment. In the dark, phy B is mainly found in the cytoplasm. Still, when induced by light, it is converted from the inactive Pr form to the active Pfr form and rapidly translocated to the nucleus, where it is segregated into subnuclear photobodies (PBs) that perform most signaling functions(Burgie et al.,

2021; Klose et al., 2020). All *A. thaliana* phy A-E are localized in photobodies, and the localization of PBs is conserved in monocotyledons and dicotyledons. In *A. thaliana*, high-intensity red-light stimulation produces large PBs with approximately 1–2 nm diameters. Conversely, in response to far-red-light stimulation, phy B is translocated into smaller photobodies or undergoes diffusion in the nucleoplasm(M. Chen et al., 2003; Van Buskirk et al., 2012).

As a plant-specific subnuclear concentrate, PB has many of the same properties as condensates formed by phase separation, such as spherical morphology, the ability to be disassembled and reorganized, and fluorescence recovery after photobleaching (FRAP) experimentally demonstrated dynamics was shown that the phy B protein is the one with the ability to assemble condensates via LLPS. Now, phy B is a light and temperature receptor for plants and can spontaneously form condensates via LLPS (Fig. 2). Moreover, the assembly of phy B PBs is highly light-dependent(D. Chen et al., 2022). The authors of this report found that the C-terminus of phy B drives LLPS through spontaneous polymerization and that its N-terminal extension (NTE) acts as a regulator in this process(D. Chen et al., 2022). Although the authors did not perform definitive in vitro analytical experiments on phy B to determine whether LLPS is intrinsic to the nature of phy B, other analyses, including FRAP, rendered 3D fluorescence images condensate fusion assays, and analysis of mutant variants, support the hypothesis that phy B condensates form through LLPS. In addition, the authors have further investigated the molecular characterization of condensate formation. As mentioned, IDRs with multivalent interactions are extensively involved in forming condensates. Analysis of phy B's NTEs using IUPred2A showed that most of his NTEs were intrinsically disordered(Burgie et al., 2021; Meszaros et al., 2018). Further OptoDroplet assays and mutant experiments further demonstrated that NTE is a phase-separated IDR that promotes LLPS(D. Chen et al., 2022). Moreover, the variant lacking NTE was insensitive to temperature elevation and, with the addition of other IDRs, still failed to restore the thermal responsiveness, suggesting the presence of sequence specificity. In addition, under the guidance of NTE, the condensate could respond effectively to temperature changes, and NTE determined the temperature-responsive properties and the physiological activities of phy B PBs. Temperature is indispensable for regulating phy B LLPS, which is significant for plant adaptive changes (D. Chen et al., 2022). This suggests that phy B PBs can play not only a role in the light-sensing mechanism but also an indispensable role in the temperature response system of plants.

Possible biological functions of PBs include serving as sites for protein degradation, acting as reservoirs for active photoreceptors, and organizing transcriptional regulation(Sheerin et al., 2015; Van Buskirk et al., 2012). Interestingly, another photoreceptor, cryptochrome (CRY) 2 PBs, activated by blue light, was also shown to be formed by LLPS, and CRYs were shown to be involved in maintaining chlorophyll homeostasis, as well as regulating the biological clock (Fig. 2A)(B. Jiang et al., 2023; X. Wang et al., 2021; J. Yang et al., 2023). CRY PBs are photoreceptors activated by blue light, and numerous experiments have demonstrated that multiple factors interact with CRY2 and phy B(Qu et al., 2023; Sanchez et al., 2020; X. Wang et al., 2021; Zuo et al., 2011). Furthermore, recent studies have shown that phys and CRYs exhibit "dark activity" in *Solanum lycopersicum* without light (J. Li and Hiltbrunner, 2021). For example, phy A could regulate carbon flux in tomato seedlings grown without light (Carlson et al., 2019). This implies that phys and CRYs may share synergistic roles in various signaling, enabling plants to cope with more complex environments.

4.1.2. Nuclear Speckles

In addition to photobody, Nuclear Speckles are another important biomolecular agglomerate that plays an important role in the regulation of gene expression. Nuclear speckles, also known as interchromatin granule clusters, are dynamic subnuclear structures enriched in pre-messenger RNA splicing factors located in the interchromatin region of the nucleoplasm of mammalian cells, which exhibit irregular punctate structures of varying size and shape. Nuclear speckles are hubs that coordinate nuclear gene expression and play essential roles in splicing

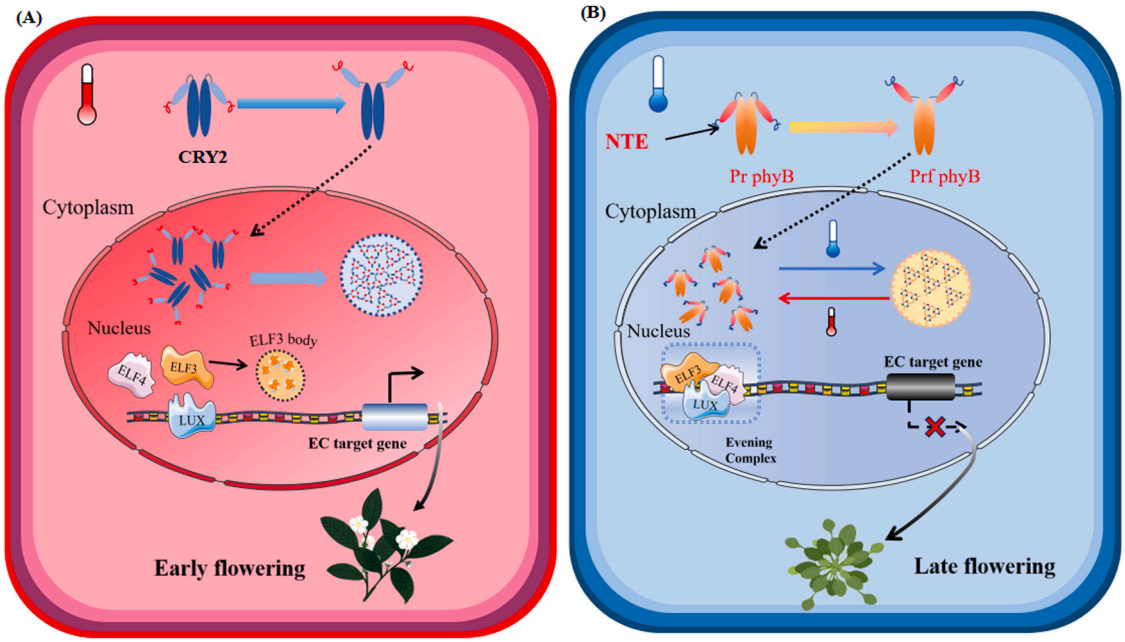

Fig. 2. Formation of ELF3 condensate under the influence of temperature and light in the nucleus and formation of CRY2 PB and phy B PB under light conditions. (A) Under low-temperature conditions, ELF3, ELF4, and LUX form the evening complex, which acts as a transcriptional repressor to inhibit the transcription of EC target genes, leading to late flowering. In response to blue light stimulation, CRY2 translocates to the nucleus to form CYR2 PB. (B) Under high temperature, the PrLD of ELF3 responds to the temperature change and forms a condensate, which leads to the disintegration of the evening complex and prevents the binding of EC to the target genes, thus allowing the expression of EC target genes and leading to early flowering. Under red light irradiation, inactive phy B was activated to form active phy B and was translocated from the cytoplasm to the nucleus to form phy B photobodies, and fusion and disassembly of phy B photobodies occurred under the influence of temperature.

regulation, RNA metabolism, and gene expression regulation (Galganski et al., 2017; Lamond and Spector, 2003; Saitoh et al., 2004). It has been demonstrated in animals that nuclear speckles, such as condensate, are formed via phase separation. Serine/arginine repetitive matrix 2 (SRRM2), a nuclear speckle marker containing multiple disordered structural domains, is vital in regulating cell splicing (S. Xu et al., 2022). It was found that the presence of SRRM2 promotes the formation and stabilization of nucleosomes, whereas the absence or reduction of SRRM2 leads to the depolymerization and destruction of nucleosomes. SRRM2 can organize condensates by phase separation and regulate the shearing process by interacting with other proteins and RNAs (S. Xu et al., 2022). Therefore, SRRM2 has a critical regulatory role in the organization and function of nucleosomes. SRRM2 has now been shown to be central and essential for organizing nuclear speckles.

In *A. thaliana*, a new flowering inhibitory gene *UBA2c* forms speckles in the cell nucleus. The UBA2 family, to which UBA2c belongs, consists of two additional members, UBA2a and UBA2b. UBA2a and UBA2b respond to abscisic acid (ABA) and form nuclear speckles, while UBA2c forms nuclear speckles with or without ABA treatment (Bove et al., 2008; Lambermon et al., 2002; Riera et al., 2006). The protein encoded by UBA2c contains two putative RNA recognition motifs (RRMs) and a PrLD. Hundreds of proteins containing PrLD have been identified in *A. thaliana*, and these PrLD-containing proteins are believed to have a high potential for phase separation (Chakrabortee et al., 2016). UBA2C regulates the major flowering repressors of *FLOWERING LOCUS M (FLM)* expression by promoting the binding of RNA polymerase II to FLM chromatin, thereby inhibiting flowering. In addition, the RRM structural domain is vital for protecting FLM from H3K27 trimethylation, a histone marker related to transcriptional repression (N. Zhao et al., 2022). The RRM and PrLD domains of UBA2c are responsible for forming nuclear speckles, which is essential for UBA2c to exert its biological function. Green fluorescent protein (GFP) experiments showed that full-length UBA2c and PrLD form speckles in vitro, while RRM does not. To determine if condensates are formed, FRAP and in vitro protein purification experiments revealed that the UBA2C-GFP signal does not diffuse in non-bleached regions, indicating that UBA2c forms gel-like condensates (N. Zhao et al., 2022). Gel-like condensates are usually formed in response to changes in temperature and pH, which are of great significance in the adaptation of plants to sudden changes in environmental stress. (J. A. Riback et al., 2017). These new findings provide direction for further understanding of plant flowering mechanisms, and therefore, further studies on the mechanism of the regulation of flowering time by the gel-like condensate produced by the structural domain-driven production of RRM and PrLD are necessary.

4.1.3. EARLY FLOWERING3 (ELF3) condensates

Plants are sensitive to changes in external conditions, especially temperature. In addition to phy B PBs, EARLY FLOWERING3 (ELF3) is involved more directly in the temperature-sensing mechanism of plants. ELF3, ELF4, and LUX ARRYTHMO (LUX) together form the evening complex (EC), which is involved in the expression of critical regulators of photosynthesis, the biological clock, phytohormone signaling, growth, and response to the environment (Andrade et al., 2022; Hutin et al., 2023). Under high-temperature conditions, ELF3 dissociates, forming condensates in a PrLD-dependent manner and causing EC to dissociate from DNA and activate target genes. At low temperatures, ELF3 condensates dissociate, diffuse in the nucleus to form EC, and inhibit transcription of target genes(Fig. 2B)(Ezer et al., 2017; Jung et al., 2020).

PrLD of *A. thaliana* ELF3 (AtELF3) is enriched in poly-glutamine (polyQ), and the length of polyQ is positively correlated with the temperature response, with PrLD exhibiting different sizes ranging from 7 to 29 glutamines in 181 natural *A. thaliana* species(Tajima et al., 2007; Undurraga et al., 2012). *Solanum tuberosum* ELF3 (StELF3) had a small PrLD, whereas *Brachypodium distachyon* ELF3 (BdELF3) had no PrLD. These plants' temperature-responsive early flowering characteristics were almost lost after substitution using StELF3 and BdELF3(Jung et al., 2020). In addition, further FRAP experiments showed that the PrLD of AtELF3 forms condensates at high temperatures via the LLPS in the nucleus and that an increase in polyQ length also results in a greater tendency to form condensates. Hutin et al. further explored the effect of the length of polyQ in ELF3 PrLD on the phase separation kinetics and showed that the LLPS of ELF3 PrLD with different polyQ lengths was sensitive to temperature and pH(Hutin et al., 2023). Furthermore, ELF3 expressed in the yeast system can lead to LLPS without other EC components, which was demonstrated by in vitro protein purification experiments.

In conclusion, AtELF3 links gene expression to temperature sensing through gene targets' protein phase-separation-mediated temperature regulation mechanism, allowing it to act as a more direct temperature receptor in *A. thaliana*. ELF3 and ELF4 can form various complexes with other proteins, such as chromatin remodeling factors, histone deacetylases, and transcription factors, thus enabling transcriptional repression at different targets(Hayes et al., 2021; H. Zhao et al., 2021). In addition, ELF3 and ELF4 are emerging as two signaling hubs integrating plant photoreception and thermoreception and their biological functions in other crops, such as *Oryza sativa*, *Zea mays*, and *Glycine max*, have also been described(Lin et al., 2021; Vicentini et al., 2023; H. Zhao et al., 2021). To cope with the food crisis caused by global population growth, adjusting the flowering time of crops according to changes in photoperiod and temperature will help to solve this challenge.

4.1.4. Guanylate-binding protein defence-activated condensates (GDACs)

During the long evolutionary process, plants have evolved various active and passive defense mechanisms to cope with pathogen infestation. The plant immune system consists of two components, PAMP-triggered immunity (PTI) and effector-triggered immunity (ETI), and these two components are interrelated(Jones and Dangl, 2006); (Ngou et al., 2022; Tsuda and Katagiri, 2010). Plants utilize several cell surface and internal immune receptors to sense various immunogenic signals associated with pathogens and subsequently rapidly activate the associated defense mechanisms, and the activation of plant resistance to these pathogens is based on the regulation of the activity of various transcription factors(Jones et al., 2016). In addition, recent studies have shown that plant guanylate-binding protein (GBP)-like GTPase superfamilies (GBPLs) play an essential role in plant defense mechanisms by assembling into plant condensates via LLPS. The authors refer to such condensates as GBPL defence-activated condensates (GDACs)(S. (Huang et al., 2021).

GBPL1 is a pseudo-GTPase that acts as an allosteric regulator, chelating catalytically active GBPL3 under basal conditions to restrict the transcription of defense genes. In response to immune signaling, GBPL3 enters the nucleus, excludes GBPL1 through the formation of GDACs, and activates the corresponding immune response.(S. (Huang et al., 2021). The role of salicylic acid (SA) in plant defense mechanisms has been extensively reviewed(Ding et al., 2022). Infected tissues in plants usually exhibit hypersensitive response (HR) responses that produce SA and activate *SYSTEMIC ACQUIRED RESISTANCE (SAR)* gene expression(Bernsdorff et al., 2016). In uninfected tissues, elevated levels of SA induce the expression of *PATHOGENESIS-RALETED (PR)* genes and improve plant disease resistance(Durrant and Dong, 2004). SA and pathogens could significantly increase the stock of GBPL3 in the nucleus, and the total amount of GBPL3 did not change throughout the cell(S. (Huang et al., 2021). In addition, droplets are produced in the nucleus as GBPL3 increases in the nucleus. This suggests that GBPL3 translocation to the nucleus and GBPL1 restriction may require immune activation. After condensate formation, GBPL1 is excluded, and within it, GBPL3 binds directly to the promoters of defense genes, recruiting specific transcriptional coactivators in the Mediator complex and RNA polymerase II, resulting in large-scale reprogramming of host gene expression to enhance disease resistance(S. (Huang et al., 2021). In addition, after removing the IDR sequence from GBPL3, condensates were no

longer formed and the plants showed decreased autoimmunity and pathogen resistance(S. (Huang et al., 2021). This implies that LLPS mediated by IDR may be an irreplaceable part of the plant defense mechanism. In addition, bioinformatics analysis of IDRs in GBPL in plants revealed the emergence of the GBPL superfamily containing IDRs at an early stage of plant evolution(S. (Huang et al., 2021). This may provide evidence for LLPS as a conserved defense mechanism in plants.

4.1.5. FCA condensates

Flowering is one of the most critical transitions in plant growth and development; therefore, the correct timing of flowering is essential in regulating the reproductive behavior of plants. Plants produce various signaling molecules to ensure that flowering occurs correctly by constantly sensing changes in external conditions(Thomson and Wellmer, 2019); (Wellmer and Riechmann, 2010). FLOWERING LOCUS C (FLC) and FRIGIDA (FRI) are two floral repressors in *A. thaliana*(Y. (Wang et al., 2020). An autonomous pathway consisting of proteins such as FCA, FPA, FY, FLD, and FLK antagonizes the expression activity of FRI on FCA, which is independent of FRI activity(Crevillén and Dean, 2011). A collection of noncoding FLC antisense transcripts, named COOLAIR, is a biologically active long-stranded non-coding RNA (lncRNA)(Whittaker and Dean, 2017). Autonomous pathway components induce alternative 3′-end processing of COOLAIR and can form an RNA-mediated chromatin silencing mechanism that regulates FLC expression. FCA is an RNA-binding protein that interacts with processing elements at the 3′-end of RNA and plays a vital role in the autonomous pathway of flowering. (P. (Li et al., 2012); (Whittaker and Dean, 2017).

Fang et al. showed that in vivo FCA can generate condensates via LLPS(Fang et al., 2019). FCA contains two N-terminal RNA-binding structural domains, a WW protein-interacting structural domain at its C-terminus, and two PrLDs(Chakrabortee et al., 2016).In addition, PrLD in FCA can undergo LLPS in vitro and the formation of condensates. However, GFP-FCA did not form droplet-like structures with phase-separation properties in vitro(Fang et al., 2019). This implies that other factors exist to regulate the formation of FCA condensate. FLX-LIKE 2 (FLL2), a PrLD-containing coiled-coil protein that can co-localize with FCA in droplets, is required for FCA function(Fang et al., 2019). Moreover, FLL2 co-occurred LLPS with PrLD of FCA. This suggests that FLL2 promotes the phase separation behavior of FCA. Previous studies of polyadenylation of FCA in the genome have not demonstrated a clear interaction of FCA with RNA 3′-end processing and polyadenylation factors in vivo(Quiroz et al., 2021). The authors of this report demonstrated by crosslinked nuclear immunoprecipitation and mass spectrometry (CLNIP-MS), a new technique developed by the authors to determine the interactors produced by FCA after formaldehyde cross-linking, that 3′-end processing factors are associated explicitly with FCA and FLL2 and confirmed that the condensate formed by FCA is a binding site for FCA and RNA 3′-end processing factors(Fang et al., 2019).In addition, the authors suggest that the FCA condensates compartmentalize the 3′-end processing factors, thereby enhancing polyadenylation of specific 3′-end processing sites(Fang et al., 2019).

Interestingly, FLL2 contains not only PrLD but also a convoluted coiled-coil domain in which the serpentine coiled-coil domain plays a role in protein interactions. The LIPRIN-α protein has the potential to oligomerize through its N-terminal twisted coiled-coil domain, and the oligomerized twisted coiled-coil domain of LIPRIN-α2 enhances the phase separation of the N-terminal fragment of the synaptic scaffold protein ELKS(Liang et al., 2021). Similarly, the coiled-coil protein Sup35 in yeast can undergo phase separation in response to pH(Franzmann et al., 2018). It indicates that the coiled-coil domain may be a potential phase separation structure.

4.1.6. TERMINATING FLOWER(TMF) condensates

During plant growth and development, plants maintain stem cell production within the shoot apical meristem (SAM) and root apical meristem (RAM) to ensure postembryonic growth and de novo formation of organs(Park et al., 2014); (Su et al., 2020). The maturation process of plant meristematic tissues undergoes a continuous differentiation from leaves to the final differentiation into flowers. Precise control of the timing of differentiation of meristematic tissues into flowers is of great significance for the adaptive reproduction of the next generation of plants, and both endogenous and external environments tightly regulate this process.

Recent studies have shown that in *Lycopersicon esculentum*, the transition time of flowers is related to the reactive oxygen species (ROS) mechanism present in the TERMINATING FLOWER (TMF) of the *A. thaliana LSH1 and Oryza G1 (ALOG)* family of genes(X. (Huang et al., 2021). In addition, ROS have been reported to play essential roles in maintaining average plant growth and improving plant resistance, such as regulating stem cell proliferation and differentiation(H. L. (Huang et al., 2019); (Weits et al., 2021). When TMF is mutated, the F-box gene *ANANTHA (AN)* is prematurely activated, leading to early meristematic tissue maturation and flowering. There is one IDR each at the N-terminus and C-terminus of TMF, termed IDR1 and IDR2, and as mentioned earlier, IDRs play a significant role in driving phase separation. The IDR1 region of TMF is enriched in asparagine. The authors mutated the eight asparagine residues in IDR1 and deleted the shorter IDR2, and the results showed a significant decrease in the condensate generation (X. (Huang et al., 2021). In addition, four cysteine residues are present in the folded region between IDR1 and IDR2, typical targets of H_2O_2-mediated oxidation. Targeted thiol modification of cysteine residues in target proteins by H_2O_2 is the primary mechanism ROS mediates biological effects in redox regulation(Cheung and Vousden, 2022); (Mittler, 2017); (Paulsen and Carroll, 2010). Further experiments showed that these four cysteine residues in TMF function as redox receptors, which sense redox conditions, form disulfide bonds, link TMF molecules, promote homodimerization of TMF, form phase-separated condensates to inhibit *AN* expression, and regulate floral transition. (X. (Huang et al., 2021). TMF interacts directly with *AN* in vivo, and *AN* is a target of TMF. Notably, both cysteine residues and IDR mutations in TMF affect its binding ability to the promoter, which regulates plant flowering time. This implies that the phase-separation ability of TMF is critical for the transcriptional binding of TMF.

Currently, most available studies on phase separation in plants have focused on the model organism *A. thaliana*, and little research has been done on crops. Huang et al. proposed phase separation in crops for the first time and modeled protein phase separation, ROS, and stem cell fate, which play critical roles in regulating plant growth and development(X. (Huang et al., 2021). Intriguingly, Chen et al. found that heavy metals with redox activity directly contribute to ROS production(Y. G. (Chen et al., 2021). Heavy metals bind directly to molecular targets, altering the cells' metabolism (Zhang and Xia, 2023). Moreover, ROS levels in plant cells can respond rapidly to environmental influences. The uptake of heavy metals into the cell by transporter proteins results in the production of excess ROS, leading to an imbalance in cellular signaling, which in turn affects the yield and quality of the final medicinal plant(H. L. (Huang et al., 2019). Therefore, ROS-promoted TMF phase separation may be an evolutionary strategy for crop adaptation to the environment, which would help guide the selection and cultivation of stress-resistant crops.

4.2. Phase separation in the cytoplasm

4.2.1. Auxin Response Factor (ARF) condensates

The regulation of phytohormones plays a decisive role in plant growth and development, including environmental adaptation, morphogenesis, stress response, and defense responses, and phytohormones induce genome-wide transcriptional programming in plants(Bao et al., 2020); (Jogawat et al., 2021); Y. (Zhao et al., 2019). Among them, transcription factors of the AUXIN RESPONSE FACTOR (ARF) family strictly regulate this process(R. (Wang and Estelle, 2014). Most ARFs present in the model organism *A. thaliana* contain three homologous

regions, N-terminal B3-type DNA-binding Domain (DBD), variable middle region (MR), C-terminal type I/II Phox and Bem1p (PB1) domain (Quiroz et al., 2021).

Recent studies have shown that ARF7 and ARF19 are mainly localized in the nuclei of root tip cells and form condensates in the response of root tissues to auxin, and have also suggested that MR and PB1 are essential in the assembly of condensates(Powers et al., 2019). The condensates formed by these two factors exhibit different properties and varying sensitivity to auxin in other parts of the root tissue; they form condensates in the nucleus of the apical tissue and are involved in the auxin regulatory mechanism. In the upper part of the root, they form condensates in the cytoplasm, which are insensitive to exogenous auxin (Fig. 3B)(Powers et al., 2019). Remarkably, an *arf19* variant that severely affects condensate formation does not form condensate in the upper part of the root and remains diffuse in the nucleus. The *arf19* variant showed auxin-responsive properties in cells that were otherwise unresponsive to auxin in wild-type plants. This implies that the ARF19 condensate has an essential regulatory role in plant roots' auxin response mechanism. In addition, FRAP experiments also showed that YFP-ARF19 from the upper part of the root has more solid-like properties. Moreover, a charged surface of the oligomerized structural domain of the mutant *arf19* C-terminal PB1 then disrupts ARF19 condensate formation in planta, which is consistent with previous findings that p62/SQSTM1 PB1 drives filament formation as well as the onset of phase separation in human cells(Ciuffa et al., 2015); (Powers et al., 2019); (Sun et al., 2018). A single-point mutation in the structural domain of PB1 directly disrupts the production of the condensate. At the same time, this does not affect the normal transcriptional function of ARF proteins, implying that the action of the condensate can be dissociated from the function of the core protein(Powers et al., 2019). In addition, strains with disrupted ARF condensates underwent significant phenotypic changes highly correlated with auxin transcriptional responsiveness, such as shortened root length, reduced cotyledon area, and increased root hair length(Powers et al., 2019).

In conclusion, the formation of ARF19 condensate can sequester the ARF19 transcription factor from contact with the outside world to prevent or inhibit auxin signaling during plant root development. This means that this distribution system of ARF proteins in the cell, driven by condensates, is vital to the mechanisms that regulate the auxin response in plant root tissues. Although ARF condensates through LLPS formation, the exact molecular mechanism remains undetermined. Further study of the influence of other factors, such as the environment, other hormones, etc., on the mechanism of ARF condensate formation is warranted.

4.2.2. Salicylic acid-induced NPR1 condensates (SINCs)

ETI is an active plant defense mechanism that induces programmed cell death against infection(Chang et al., 2022). To keep programmed cell death tightly contained to the infected region without spreading, SA and NONEXPRESSOR OF PATHOGENESIS-RELATED GENES1 (NPR1) cause uninfected cells to acquire SAR(Chang et al., 2022); (Rodriguez et al., 2016). In addition, NPR1 plays a crucial role in plant defense by regulating the interaction of SA with other plant defense signals and phytohormones. The roles and mechanisms of NPR1 as a gene regulator in plant defense and stress response have been extensively described(Jin et al., 2018); (Lai et al., 2018); (Olate et al., 2018); Raul (Zavaliev and Dong, 2023). Recent reports have shown that NPR1 contains IDRs and undergoes a phase transition in the cytoplasm in response to SA to form salicylic acid-induced NPR1 condensates (SINCs)(R. (Zavaliev et al., 2020). To clarify the role of IDRs in SINCs, the authors performed mutation assays on the cysteine clusters of the three redox-sensitive IDRs present in NPR1: *rdr1* and *rdr2* formed agglutinate in the absence of SA induction, whereas *rdr3* was unresponsive to SA induction and did not form in either the nucleus or cytoplasm(R. (Zavaliev et al., 2020). This suggests that cysteine clusters in RDRs play a crucial role in responding to SA signaling and forming condensates through phase transitions. Further, the authors performed a proteomic analysis of SINC. They showed that positive regulators of ETI, proteins involved in the ubiquitination mechanism, and other proteins related to plant defense and stress were enriched in SINC(R. (Zavaliev et al., 2020). NPR1 recruits and specifically binds Cullin 3 RING E3 (CRL3) ligase to SINC, thereby ubiquitinating ETI-positive regulators localized in SINC, such as the master immune regulator EDS1 and the transcription factors WRKY54 and WRKY70(J. (Chen et al., 2017); R. (Zavaliev et al., 2020). This suggests that SINC slows ETI-induced programmed death with elevated SA levels through targeted degradation, enhancing plant survival during infection

Intriguingly, with the deepening of LLPS research, the evidence for the involvement of LLPS in plant defense mechanisms is increasing (Ngou et al., 2022); Raul (Zavaliev and Dong, 2023); R. (Zavaliev et al., 2020). Unlike NPR1, GBPL3 also acts as the core of the condensate, promotes condensate formation, and recruits plant disease resistance-associated factors into the condensate to exert the corresponding disease resistance regulatory capacity. Similarly, in both condensates, IDRs have irreplaceable roles in condensate formation. As many translation-related factors are present in both GDACs and SINCs, does this imply that the condensate formed by phase separation plays a vital role in determining whether programmed cell death occurs. It makes us wonder whether phase separation is an ancient evolutionary mechanism for plant defense against pathogens.

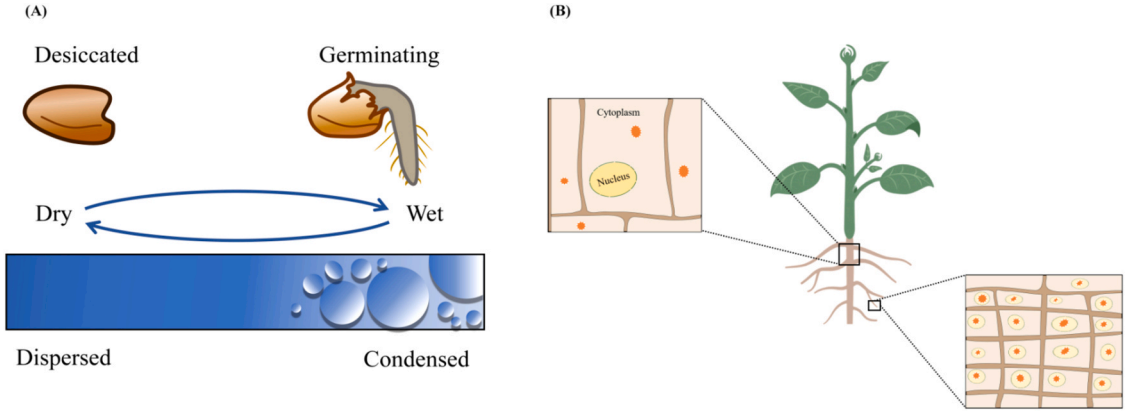

Fig. 3. Plant cytoplasmic condensates are involved in water sensing as well as seed germination. (A) FLOE condensate is engaged in the water-sensing mechanism of seeds. It can cycle between water uptake and dehydration before the seed has germinated. (B) Cells in the upper root portion stop growing, and ARF condensates localize in the cytoplasm in the upper root. In the root tip, ARF condensates localize in the nucleus and participate in the apical growth hormone regulatory mechanism, attenuating the growth hormone response in the root tip.

4.2.3. FLOE condensates

Plants have produced seeds over evolution to protect the offspring from germinating and create a new generation under the right conditions. The germination of a seed depends on various conditions, such as temperature, plant hormones, salinity, hydration, etc., to ensure that it sprouts at the most appropriate time(Arif et al., 2020); (Geng et al., 2013); (Parwez et al., 2022); (Zhou et al., 2020). If necessary, the seed remains viable for thousands of years, even in harsh environments (Sallon et al., 2020). The seeds can wait for the right time until they germinate; once they do, they lose their drought tolerance(Rajjou et al., 2012). Hydration is a critical step in seed germination, but the exact mechanism by which seeds germinate in response to water remains unclear.

Recent studies have shown that the prion structural domain-like protein FLOE1 in *A. thaliana* can segregate by phase separation, enabling plant embryos to sense water stress (Fig. 3A)(Dorone et al., 2021). The authors identified the presence of the structural domain of PrLD in 14 of the proteins enriched in the seeds by transcriptome analysis. Previously, the role of PrLD in driving phase separation has been widely demonstrated(Alberti et al., 2019). There are two distinct disordered regions in FOLE1: one enriched for aspartic acid and serine (DS-rich), the other enriched for glutamine, proline, and serine (QPS-rich). Regularly spaced tyrosine residues are present in the QPS PrLD. Increasing the number of tyrosine residues, i.e., elevated multivalency, induces a shift in the droplet-like agglutination toward a gel-like consistency (Dorone et al., 2021). Functional analysis of the two disordered regions showed that deletion of the QPS PrLD caused a decrease in the formation of FLOE1 condensates both in vivo and in vitro; conversely, deletion of the N-terminal DS PrLD caused a shift of FLOE1 toward a gelatinized state and, if the serine in the DS PrLD was replaced, the same effect as DS PrLD deletion was observed(Dorone et al., 2021). The fact that serine and tyrosine are both aromatic amino acids suggests that aromatic amino acid residues may have opposite roles in DS PrLD and QPS PrLD. FLOE1 can undergo multiple hydration-dehydration cycles and form condensates in response to changes in water potential by water uptake in the plant embryo. Mutant experiments have shown that *floe1* mutants increase seed germination under drought-stress conditions. In contrast, they behave like the wild type under normal conditions, suggesting that FLOE1 can prevent seed germination under drought-stress conditions(Dorone et al., 2021). The opposite phase behaviors mediated by the two disordered structural domains of FLOE1 are both hydration-dependent, which allows FLOE1 to sense changes in water potential to regulate seed germination. It remains to be investigated whether FLOE1 directly feels moisture changes and whether other factors are involved in condensate formation.

Furthermore, the phased segregating phenotypes were highly diverse, as analyzed by FLOE homologs in seed plants. Such breadth suggests that the *FLOE* gene family may play a role in plant functional phenotypic variation(Dorone et al., 2021). This indicates that the condensate formed by FLOE1 hydration may be a putative water-sensing mechanism that prevents seed germination under unsuitable conditions and improves the probability of plant survival under stress conditions, which also provides solution ideas for global crop response to drought stress.

4.3. Phase separation in organelle

4.3.1. STT condensates

LLPS has now been shown to be involved in protein sorting in chloroplasts(Ouyang et al., 2020). Most chloroplast proteins are synthesized as precursors in the cytoplasm and subsequently translocated into the chloroplasts (Lee et al., 2017). These precursors may need to cross three membrane systems before they are finally localized to their destination. Therefore, correct protein localization and sorting are essential. Two membrane-bound transport systems aid in translocating proteins in vesicles: the chloroplast secretory (cpSec) pathway and the chloroplast twin arginine transport (cpTat) pathway. In particular, the cpTat pathway consists of three proteins, Tha4, Hcf106, and cpTatC, which transfer folded proteins into the lumen. The CpTat pathway substrates have transit peptides that contain stromal (STP) and lumen-targeting peptide (LTP)(Lo and Theg, 2012); (Ouyang et al., 2020).

Recent studies have identified and characterized two anchoring proteins, STT1and STT2, involved in protein sorting of the cpTAT pathway to the vesicle-like membrane via LLPS(Ouyang et al., 2020). Yeast two-hybrid experiments showed a strong interaction between STT1/STT2 and OE23, a substrate of the cpTAT pathway. However, the binding of the substrate to the vesicle-like membrane was significantly increased only when both STT1 and STT2 were present. Bimolecular fluorescence complementation experiments (BiFC) demonstrated the existence of a direct interaction between STT1/STT2 and the presence of an STT1-STT2 complex in plants. STT1 and STT2 contain an N-terminal IDR as well as a C-terminal structural domain with a repeating sequence of five anchoring protein repeats, and STT1/STT2 form a heterodimer based on the C-terminal structural domain, and the IDR of the N-terminal structural domain is essential in the interaction with the signaling peptide of the cpTat pathway. When the STT complex is mixed with OE23 LTP in vitro up to a specific concentration, the solution shifts to a milky white color, and droplet-like spheres of varying sizes are formed, which is one of the characteristics of LLPS(Ouyang et al., 2020). This implies that the cpTat signaling peptide may specifically induce phase separation of STT complexes in vitro. In live cell chloroplasts, the fluorescent signal of STT1 or STT2 alone, when co-expressed with OE23, does not recover after photobleaching; however, when STT complex is co-expressed with OE23, it rapidly recovers and undergoes rapid exchange with those around it(Ouyang et al., 2020). These results suggest that LLPS occurs in cohesions formed by STT in complex with cpTAT substrates in living cells and that droplet-like structures that isolate inclusions from the outside world may be a mechanism to protect hydrophobic structures from degradation. The ability of some *stt1* and *stt2* variants to undergo LLPS in vitro is impaired, and when these variants are expressed in plants, their ability to promote cpTat substrate translocation is diminished(Ouyang et al., 2020). This result is consistent with the conjecture that the phase separation of STT1 and STT2 is essential in the cargo sorting mechanism in the cpTat pathway. In addition, Hcf106, one of the components of cpTat, was shown to be a binding mate for STT1-STT2 interactions. The addition of Hcf106 to the mixture of STT complex and OE23 in vitro inhibits LLPS, implying that there may be competition between OE23 and Hcf106; in vivo, it can alter the occurrence of cargo translocation in the cpTat pathway (Fig. 4A) (Ouyang et al., 2020). However, it remains unclear whether the LLPS driven by STTs is subject to changes in physicochemical conditions within the chloroplast and whether the PTM affects the occurrence of this process(Zheng et al., 2021). In addition, SecA and SecA2 in the cpSec pathway, cpSRP 43 and cpSRP 54 in the cpSRP pathway, as well as PEX 5 and PEX 15, which are present in the peroxisomal protein translocation machinery, likewise have IDRs(Ouyang et al., 2020). This implies that LLPS may be a generalized protein-sorting mechanism in plants. In summary, the authors of this report provide a model for the LLPS-mediated protein sorting pathway in plant chloroplasts.

4.3.2. Pyrenoid matrix

Pyrenoids are found in the chloroplasts of a wide range of photosynthetic eukaryotes, and ribulose-1,5-bisphosphate carboxylase/oxygenase (Rubisco) present in the substrate of pyrenoids and plays a critically important role in CO_2 fixation, as well as in facilitating photosynthesis, and mediates 30 % of global carbon fixation(Rodriguez et al., 2016). Rubisco acts as a bifunctional enzyme that catalyzes the carboxylation and oxidation of ribulose-1–5 bisphosphate (RuBP) and limits the efficiency of photosynthesis(Wei et al., 2024). Photosynthetic microorganisms have evolved a CO_2-concentrating mechanism (CCM) in response to this challenge, and the CCM sequesters Rub and provides a

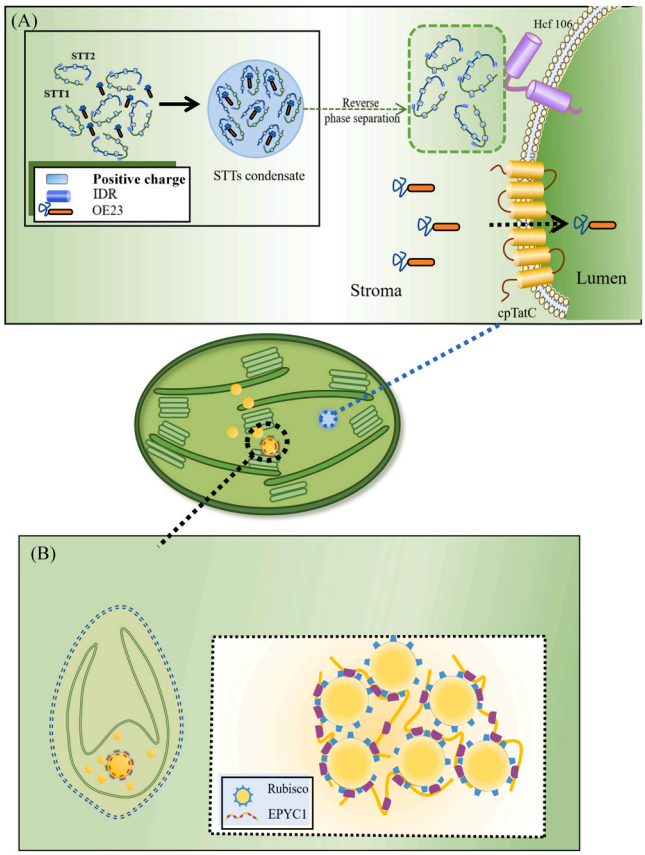

Fig. 4. LLPS that occurs in chloroplasts. (A) LLPS is involved in protein sorting in chloroplasts. STT1 interacts with STT2 via the C-terminal anchoring protein structural domain to form a heterodimer and, via LLPS, forms a condensate with the cpTat pathway substrate, OE23. STT condensate crosses the vesicle membrane and binds to the cpTat transporter system, Hcf106, where it undergoes an electrostatic interaction that results in a reverse phase separation and the passage of OE23 across the vesicle membrane into the lumen. (B) Condensates are formed between EPYC1 and Rubisco through multivalent interactions to enhance CO_2 fixation and photosynthetic efficiency.

high concentration of CO_2, thereby suppressing the losses incurred in photorespiration(Bracher et al., 2017); (Hennacy and Jonikas, 2020); (Rae et al., 2017). Previous studies have shown that the arrangement and structure of the pyrenoid matrix vary in different species, either in the form of densely arranged crystals or in an amorphous form. Furthermore, other species produce pyrenoids differently for pyrenoids, with some species growing them by fission and some species having them de novo(Bertagnolli and Nadakavukaren, 1970); (Holdsworth, 1968).

The pyrenoid matrix in *Chlamydomonas reinhardtii* has recently been shown to be a droplet-like condensate formed by phase separation (Rosenzweig et al., 2017); (Wunder et al., 2018). FRAP showed that mixing of the pyrenoid matrix occurs on a time scale of about 20 seconds, and cryo-electron tomography (cryo-ET) analysis of the pyrenoids further revealed that their shape is droplet-like spherical and that smaller spheres can fuse to form larger spheres(Rosenzweig et al., 2017). In addition, the pyrenoid matrix dissolves into chloroplasts during cell division and in response to high carbon dioxide concentrations, exhibiting fluid-like behavior(Rosenzweig et al., 2017). ESSENTIAL PYRENOID COMPONENT (EPYC)1 is a protein necessary for the localization of Rubisco in the pyrenoid nuclear matrix. It may act as a scaffold in the condensate (Fig. 4B)(Mackinder et al., 2016); (Rosenzweig et al., 2017). In this report, the authors argue, based on the presence of four highly similar repeat sequences in EPYC1, that there are four binding sites to Rubisco and propose that the aggregation effect is suppressed if the number of binding sites on Rubisco is an integer multiple of the number of binding sites on EPYC1, which the authors refer to as the "Magic Number" Effect.(Rosenzweig et al., 2017). In vitro experiments have demonstrated that purified Rubisco in contact with EPYC1 rapidly forms droplets through phase separation, is unlimited in size, and may even be much larger than the pyrenoid(Wunder et al., 2018). However, Wunder et al. showed that single-repeat EPYC1 can also unmix with Rubisco and that *epyc1* variants containing 3–5 repeats can deposit Rubisco similarly to EPYC1(Wunder et al., 2018). This suggests that EPYC1 may have more than the putative four binding sites, implying that the "Magic Number" Effect may not apply to the interaction between EPYC1 and Rubscio. In addition, Shan He et al. showed that the Rubisco holoenzyme contains eight EPYC1 binding sites and that EPYC1 binds to Rubisco through salt bridges and hydrophobic interactions(Shan (He et al., 2020).

Although many photosynthetic species have evolved CCM mechanisms to improve the efficiency of CO_2 fixation and photosynthesis, most C_3 plants are still deficient in active CCMs(Araus et al., 2002). Therefore, introducing functional heterologous CCMs into important crops through bioengineering is a strategic measure to mitigate the challenges posed by global population growth in agriculture. Existing models have made some progress in constructing functional CCMs, such as the recombinant Rubisco-EPYC1 condensate in *A. thaliana.*(Adler et al., 2022; Atkinson et al., 2020). Moreover, reducing the efficiency of photorespiration by increasing the concentration of CO_2 at the Rubisco site is also considered to have an excellent potential for yield increase(Weber and Bar-Even, 2019). This creates far-reaching prospects for increasing terrestrial crop yields by utilizing bioengineering to address the challenges posed by global agricultural dilemmas.

5. Conclusion and prospect challenges

LLPS concentrates multivalent biomolecules when needed to classify and generate appropriate cellular responses. Research on phase separation and molecular condensates has rapidly matured over the past decade. Phase separation in the context of plant biology is now widespread, yet there is little research on the specific mechanisms by which it occurs. Future research needs to focus on how to analyze the biological function of phase separation and whether phase separation is an essential mechanism in plants.

There is still a great deal of confusion about the mechanisms by which biocondensates interact with the endomembrane system in plants, and this will be a hot topic in the coming years. While we observe no other membrane-bound structures in the nucleus and only biocondensates or the entire dynamic compartment of chromosomal territories, in the cytoplasm, we have both biocondensates and membrane-bound structures (Li and Pang, 2023; Kusumaatmaja et al., 2021). Furthermore, at the plasma membrane, the IDR of SFH8 mediates LLPS, the phase separation of SFH8 enables the translocation of polar proteins, and SFH 8 condensate may affect lipid aggregation. Also, the IDR of SFH8 is evolutionarily conserved. This work provides a port of condensate that interacts with membranes and provides more favorable evidence that phase separation is an ancient evolutionary mechanism. We believe such special condensates may be discovered more often in the following work. In addition, some questions about phase separation still need to be solved. In particular, the size and number of condensates formed and their location. Are there some common mechanisms or components among bioconcentrates? How do plants control the occurrence of phase separation in response to rapidly changing environments? What magical reactions can occur with condensates that interact with membranes? Are there differences between biocondensates formed in different parts of the cell? Research in this area requires a multidisciplinary approach, and we believe that the next decade will bring many more exciting discoveries in this open field.

CRediT authorship contribution statement

Pengguo Xia: Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Yang Huang:** Writing – original draft, Validation, Resources, Project administration, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Contributions

P.X. came up with the idea for this article. Y.H. collected and wrote the paper. P.X. put forward the guidance for the revision of the article.

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